

MIE 402 ME Lab II, Spring 2026

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LAB 1: Dynamic data sampling and frequency analysis, Feb. 10-13

Objectives

- Become familiar with digital data acquisition, hardware, and software
- Explore common data acquisition pitfalls and possible remedies
- Investigating the sampling in time-domain
- Investigating the sampling in frequency domain
- Understanding the effect of sample rate and number of samples
- Understanding FFT including aliasing and filtering

Equipment

- Moku:Go as a DAQ hardware
- Microphone and speaker
- Sound creating instruments
- MATLAB data acquisition script

Task

1. Read the “Data Collection Procedure for Sampling Lab” (page 2-3).
2. Connect the microphone to the input of DAQ
3. Acquire different sound signals with different sampling frequencies
 - a. Record at least 3 different sounds (tuning fork, finger snapping, whistling ...)
 - b. For each sound, sampling with $100\times f$, $10\times f$, $3\times f$, $1.4\times f$, and $1\times f$ for ≥ 1 sec.
4. Replay the sound through the speaker to see whether they are distinguishable
5. Save data (.mat files) so that you can do analysis
6. Show the GTA one valid data set (e.g., $f_s = 100\times f$) and a corresponding frequency-domain spectrum (Pre-lab 1) with a clear peak matching the tuning fork frequency.

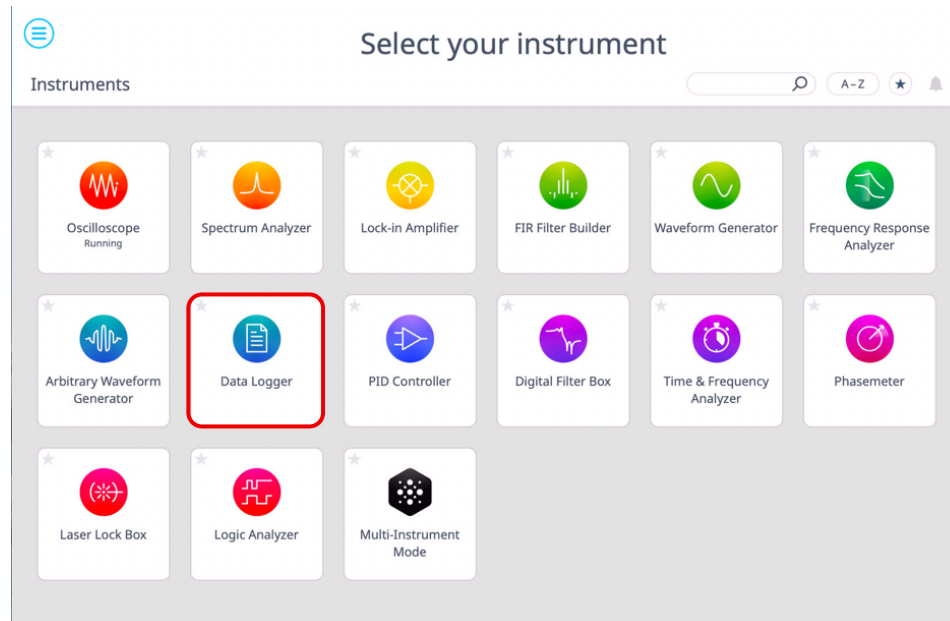
Deliverables in your report

- Compare the average, rms measurements against the theoretical frequency
- Compare different sampling frequencies and their effect on acquired signals
 - Compare the signals in terms of their time-domain waveforms
 - Compare the signals in terms of their frequency spectra
 - Compare the signals in terms of the sound replayed through the speaker

Data Collection Procedure for Sampling Lab

Connecting the DAQ and microphone:

- Connect the microphone to one of the input channels of the Moku:Go (i.e., the two BNC connectors on the left)
- Start the Moku APP on the desktop
- Select the Moku with the matching serial number. Select “Data Logger”

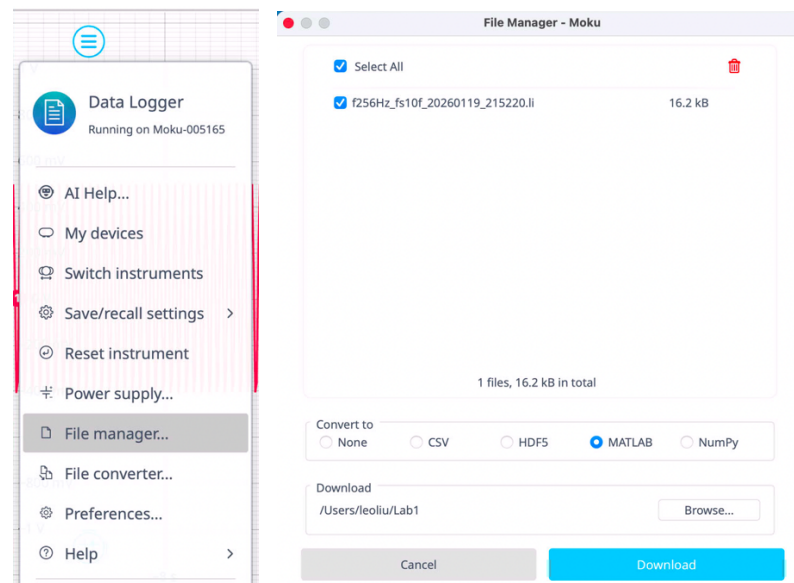


Collecting data:

- Pick the tuning fork and find its fundamental frequency f on it.
- In the Data Logger, enter the sampling frequency under “Acquisition” as “Rate”. Choose acquisition mode, such as “Precision”.
- Activate the Input channel by toggle the on slider to the right (e.g., Input 1 if you connect the microphone to the leftmost BNC connector). Deactivate the Input channel you don’t use.
- In “Logging”, choose “Immediate” to record the data right after clicking the RED record button or choose “Delayed” and set the delay time accordingly.
- In “Logging”, enter the recording length under “Duration” (ensuring ≥ 1 sec).
- Enter the prefix for the name of the recorded data file under “Filename prefix”, such as “f256Hz_fs10f” for using the 256 Hz tuning fork and try to sample the data at 2560 Hz ($f_s = 10f$).
- Click the RED record button under the waveform to start recording.



- Click the  button (or  button and choose “File manager...””) to open the “File Manager - Moku” window.
- Select the data file(s) and choose the “Convert to” “MATLAB”. Browse the target folder and then click “Download” to get your data in “.mat” format.



- Plot data and replay the sound using the provided “LAB1_PlotSignal_PlaySound.m”. After validating successful recording of meaningful data, copy all your data files to your flash drive so that you can analyze them for preparing your lab report.